## MC 451 Inferencing Data in R rubric

**100 points.** Due Friday, December 4, 11:59 PM.

**How this is scored.** Add the points from each row; the total is out of 100. A first attempt is graded forgivingly: a part that is there but incomplete earns partial credit, and 0 is kept for a part that is missing. Length: 400 to 700 words of your own writing (hypotheses, the four reports, interpretation and the limits paragraph), not counting code or output. Word counts cover your own writing, not the title, headings, tables, references, figure captions, code, or output, and writing more than the range never costs points. Scores on work that shows effort are adjusted with the course's curve: your final percentage is 25 percent plus 75 percent of your rubric percentage, so 50 of 100 on the rubric becomes 62.5. Work at two-thirds or more of the minimum length counts as showing effort. Submissions are unlimited before the deadline, so you can revise and submit again.

- The worksheet is 13-inferencing-data-mc451.qmd. Its settings block gives `group2` your hand-coded variable, so one of the four tests is on the thirty messages you coded; the calls to fill are the tests and their effect sizes. The Inferencing Data risks guide shows the traps on this job and how to check for each.
- Run all four tests: a chi-square test of independence with `chisq.test()` on an `xtabs()` table, tidied with `broom::tidy()`, and Cramér's V from `effectsize::cramers_v()`; an independent t-test with `t.test()` (Welch's version) tidied with `broom::tidy()`, and Cohen's d from `effectsize::cohens_d()`; a one-way ANOVA with `aov()` and `summary()`; and a linear regression with `lm()` and `summary()`. `v2v::run_chi_square()` and `v2v::run_t_test()` print the same numbers in one labeled block and belong after the plain calls, as a check.
- The worksheet names the four pairs of variables, one of them your hand-coded variable against message length; the variable types decide the test, and naming it with its reason is your part.
- The Test group of the R Companion's Write R code writes each test in this form, the plain call first and the v2v check after it, for the settings-block names; its Check my test choice reports whether a test you have already chosen fits your variable types, and a fitting test earns the Test choice points only with your own reason beside it; its APA numbers to check your sentence prints a test's numbers in APA form so you can check the sentence you wrote, and the four reports and the interpretation are your own wording.
- Every script opens with a settings block: load the data, rename the variables you use to outcome, `group2`, `group3`, predictor, `cat_a`, and `cat_b`, and make the categorical ones factors. The code below the block uses only those names. This is what lets the same script run on your own data for the White Paper.

- How this loses points: reporting p and stopping. The effect sizes are worth twice the execution here, on purpose: a p-value says something happened, and only the effect size says whether it matters.
- What it feeds: your White Paper, which runs this same script on your own variables after you change its settings block. The Twitch results stay here; the paper reports only what your own data shows, and only the tests your questions call for.

*Inferencing Data in R, 100 points*

| Criterion | Exemplary | Proficient | Developing | Beginning |
|---|---|---|---|---|
| **Hypotheses**<br>8 points | **8 points.** Four hypotheses in plain English, written in Part 0 before any code, one for each of the worksheet's four pairs of variables, each naming its variables and the relationship you expect. | **6 to 7 points.** Four stated before the code, but one variable is vague, or one hypothesis is a topic instead of a relationship. | **4 to 5 points.** Fewer than four, or stated after the results. | **0 to 3 points.** Missing. |
| **Test choice with its reason**<br>12 points | **12 points.** Each hypothesis takes the test its variable types call for (two categorical: chi-square test of independence; a two-group category against a continuous variable: independent t-test; a category with three or more groups against a continuous variable: one-way ANOVA; a continuous outcome and a continuous predictor: linear regression), with one sentence on why that test and not another. | **9 to 11 points.** The right four tests, but a reason is missing or would fit any test. | **6 to 8 points.** A test that does not fit its hypothesis's variable types, with a reason given. | **0 to 5 points.** No tests chosen, or tests that do not fit with no reason. |

| Criterion | Exemplary | Proficient | Developing | Beginning |
|---|---|---|---|---|
| **Execution, including the settings block**<br>12 points | **12 points.** All four tests run in the worksheet on your analysis dataset, the output shown is what the code produces, and the settings block gives `group2` your hand-coded variable, with the code below it using only the names the block sets. | **9 to 11 points.** All four run, but `group2` is not your hand-coded variable or the code below the block names the Twitch columns directly, or one test runs on a different variable than its hypothesis names with no word on why. | **6 to 8 points.** The code stops with an error, or the output is pasted in instead of produced. | **0 to 5 points.** Not run. |
| **APA reporting**<br>20 points | **20 points.** Each of the four reported in APA format with the test statistic, the degrees of freedom, the p-value, and the n. | **15 to 19 points.** One of the four pieces is missing on one or more tests, most often the n or the degrees of freedom. | **10 to 14 points.** The statistic and p only, or numbers copied from the console without APA form. | **0 to 9 points.** Not reported. |
| **Effect sizes**<br>24 points | **24 points.** The right statistic for each test (Cramér's V, Cohen's d, eta-squared computed from the sums of squares, and R-squared with the slope in your outcome's units), each with what it means in the units of the data rather than only as a label such as "small". | **18 to 23 points.** All four reported with their labels, but one has no translation into what it means for the data. | **12 to 17 points.** Computed but not explained, or the wrong measure for a test. | **0 to 11 points.** Missing: reporting p and stopping. A p-value says something happened; only the effect size says whether it matters. |
| **Interpretation**<br>16 points | **16 points.** Two to three sentences in plain English after each test, saying what the result means for its hypothesis. | **12 to 15 points.** Plain English, but it restates the numbers without saying what they mean for the hypothesis. | **8 to 11 points.** One sentence each, which usually means the "so what" is missing, or a reading that treats a small p as a large effect. | **0 to 7 points.** None. |

| Criterion | Exemplary | Proficient | Developing | Beginning |
|---|---|---|---|---|
| **What the results do not license**<br>8 points | **8 points.** One paragraph: a significant test on this sample is not a claim about Twitch, and a variable that belongs to a channel (its game, its audience) varies across about fifty channels, not tens of thousands of messages. | **6 to 7 points.** A limit, but it covers only one of the two points. | **4 to 5 points.** A generic limit ("more research is needed"), or a limit of the data in general rather than of these results. | **0 to 3 points.** Missing. |

### Before you submit

Inferential Check recomputes a chi-square or a t test from your own table or group figures and returns the statistic, the degrees of freedom, p, the effect size, the assumptions your data may be breaking, and the APA sentence to compare against what you wrote. It is a second opinion on your R output, not a replacement for it. It does not grade, and nothing you enter leaves your machine. It does not cover ANOVA or regression, so check those two against the worked examples in Chapter 13.